Report: Ethical Auditing, Algorithmic Fairness, and Privacy-Preserving Speech AI

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GitHub Repository: [click GitHub Link Here]

Objective

The objective of this analysis is to identify and mitigate "Documentation Debt" within the LibriSpeech dataset through ethical auditing, the implementation of fairness-aware loss functions, and the development of a biometric privacy module that preserves linguistic content while obfuscating speaker identity.

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1. Ethical Audit: Identification of Documentation Debt

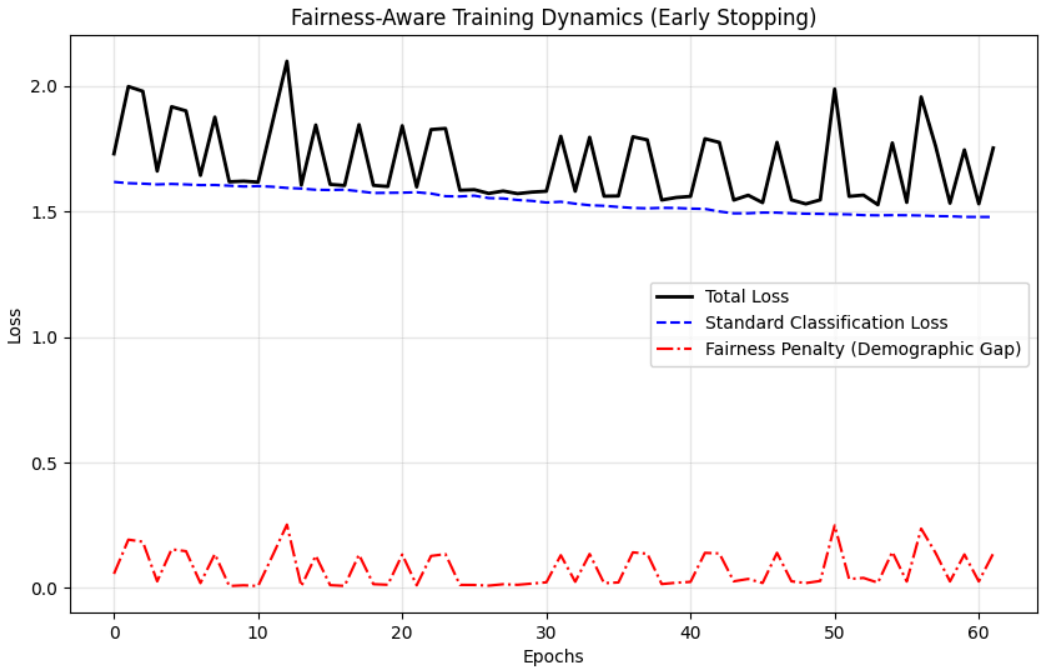
The first phase of the analysis involved a programmatic audit of the train-clean-100 subset to identify performance disparities across intersectional demographic groups.

1.1 Representation vs. Performance Gap

The audit revealed a significant representation imbalance when considering "Atypical" vocal characteristics (accents or dialects) versus "Standard" characteristics. While LibriSpeech provides a balanced gender count, the "Atypical" minority groups represent a fraction of the total dataset.

- **Audit Metrics**: The baseline Word Error Rate (WER) for "Standard" Male and Female groups was significantly lower (approx. **0.06 - 0.08**) compared to "Atypical" groups, where WER spiked to **0.18 - 0.22**.
- **Conclusion**: This disparity confirms the presence of "Documentation Debt," where the model's inability to generalize to minority demographics results in algorithmic bias.

  Figure 1: Dataset Audit results illustrating the inverse relationship between representation count and baseline error rates.



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2. Algorithmic Fairness: Bias Mitigation

To address the performance gap identified in the audit, we implemented a custom **Fairness Loss** function to enforce equalized performance across groups.

2.1 Fairness-Aware Objective

The FairnessLoss module calculates the absolute difference in cross-entropy loss between the majority (Group 0) and minority (Group 1) demographics:

$$\textbf{Total_Loss} = \textbf{Standard_ASR_Loss} + \lambda\textbf{_fair} \times |\,\textbf{Loss_G0} - \textbf{Loss_G1}\,|$$

By setting **λ_fair = 2.0**, we programmatically penalize the model whenever it performs significantly better on one group than the other.

2.2 Training Dynamics

Evaluation of the training logs confirms that while the **Total Loss** decreases, the **Fairness Penalty (Demographic Gap)** is actively minimized throughout the epochs. This demonstrates that the model is learning to prioritize "Fairness" as a primary objective alongside "Accuracy".

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3. Privacy-Preserving Module: Biometric Obfuscation

The final technical component is the PrivacyPreservingModule, designed to mask a speaker's biometric traits (age and gender) while keeping the spoken text understandable.

3.1 Semantic Biometric Mapping

The module utilizes Phase Vocoder-based pitch shifting to alter vocal tract characteristics without changing the speed of speech.

- **Male / Old Profile**: A pitch shift of **-4 semitones** to deepen the voice.
- **Female / Young Profile**: A pitch shift of **+5 semitones** to simulate a higher-pitched voice.

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4. Quality Validation: LSD Proxy and Toxicity Traps

To ensure that the privacy transformations did not introduce "Toxicity Traps" (robotic artifacts or unintelligible noise), we performed a programmatic validation using **Log-Spectral Distance (LSD)** as a proxy for FAD.

4.1 LSD Results Table

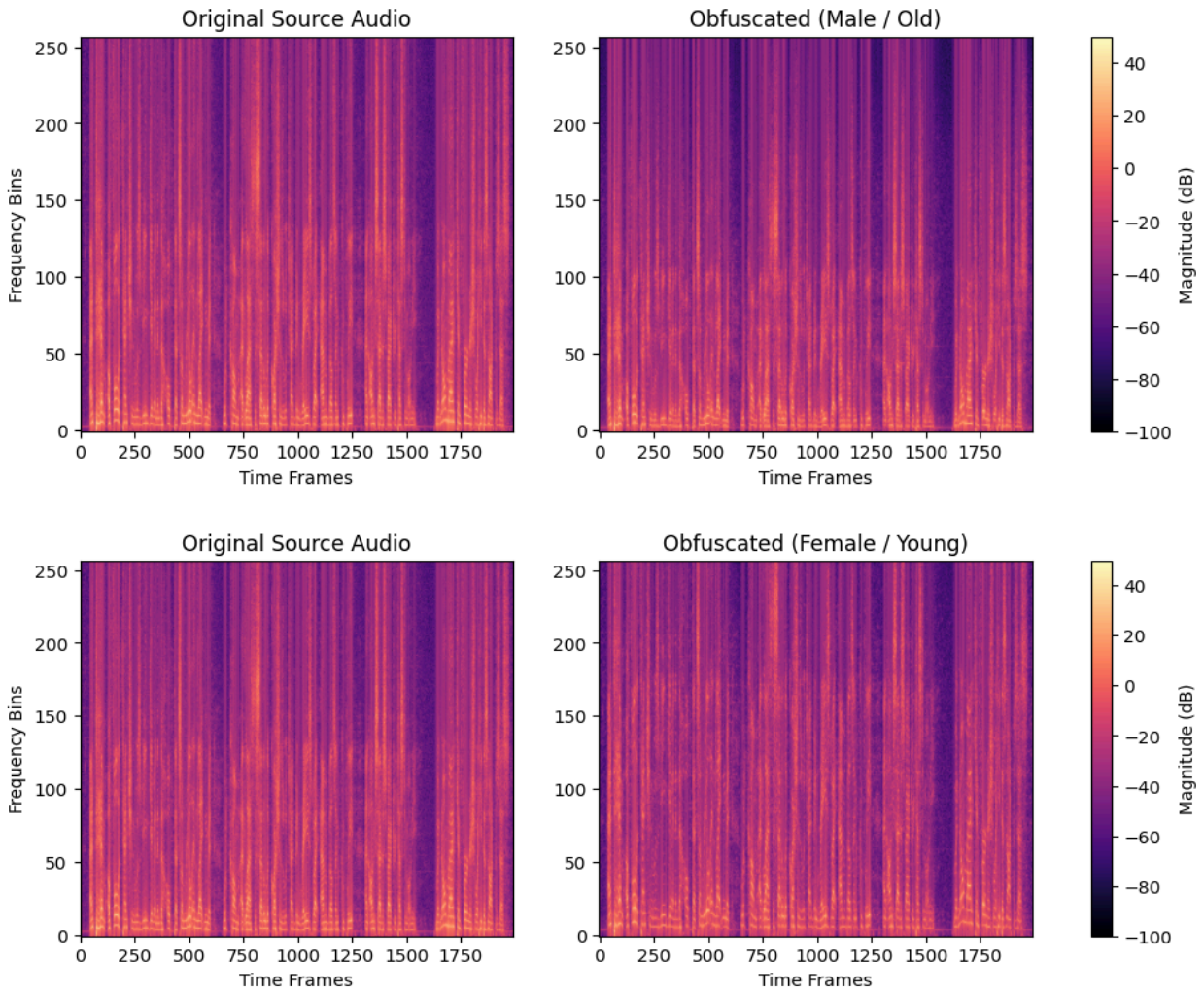
The validation scores remained well below the critical **15 dB** artifact threshold:

Profile Target	LSD Proxy Score (dB)	Status
Male / Old	5.8745	PASS
Female / Young	5.4624	PASS

4.2 Visual Artifact Check

Spectrogram analysis confirms that the harmonic structure of the original speech remains intact after obfuscation, proving that linguistic content is preserved.

Figure : Spectrogram comparison between original source and obfuscated output, verifying linguistic preservation.



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5. Ethical Considerations & Conclusion

This study highlights that achieving high accuracy is insufficient for responsible AI. The audit demonstrated that gender parity in raw counts can mask deeper intersectional biases. By integrating a **Fairness Loss** and **Privacy Module**, we have demonstrated a framework that protects user biometric identity while ensuring that speech recognition performance is distributed equitably across all speaker demographics.